

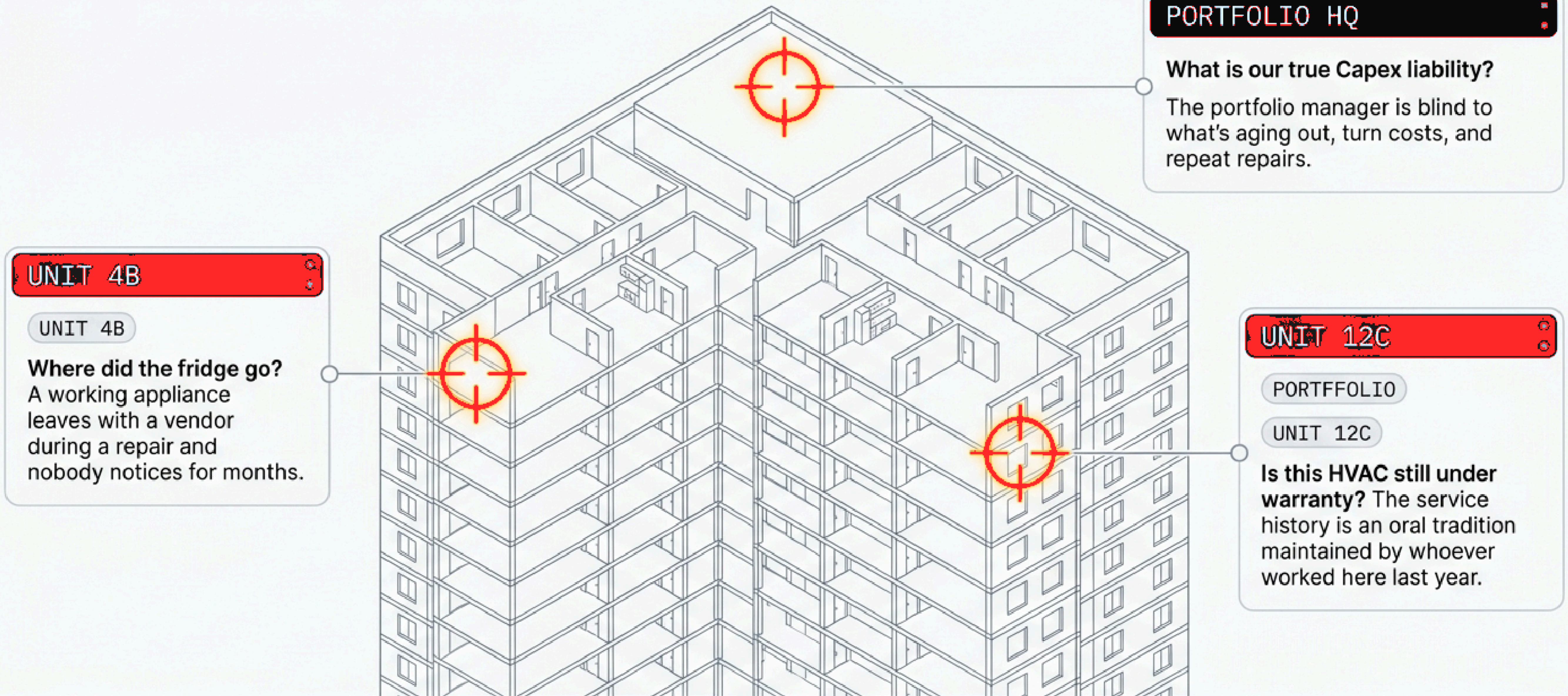
● NAMEPLATE ASSET INFRASTRUCTURE



Nameplate: Every appliance accounted for.

The immutable asset record & physical chain-of-custody engine for multifamily portfolios.

- **HARDWARE-LINKED LEDGER**
- TAMPER-EVIDENT NPID
- OFFLINE-FIRST SYNCHRONIZATION
- CAPEX RECOVERY ENGINE

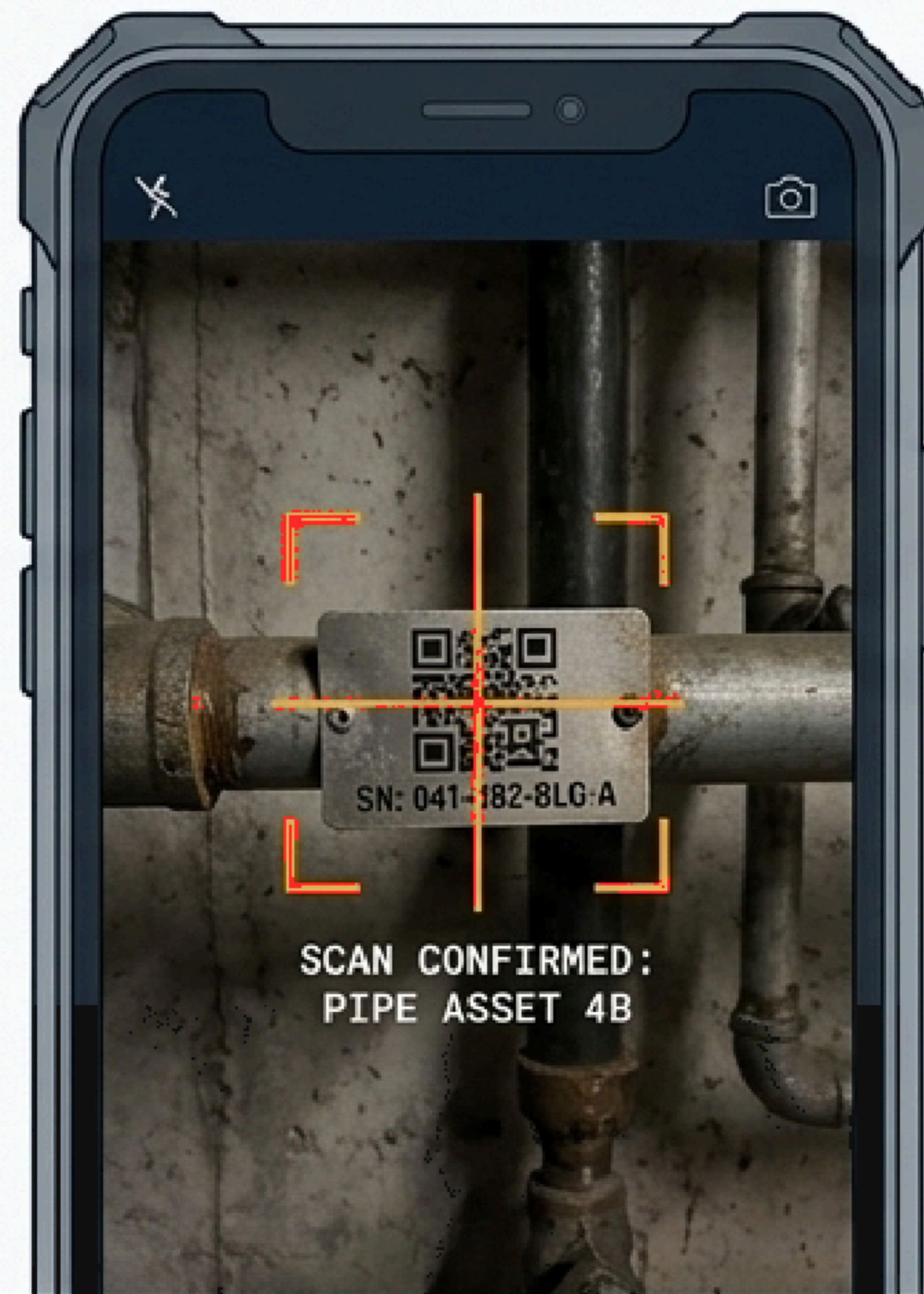


Spreadsheets decay. When an asset registry rots into fiction, operators are forced to make \$10,000 capital expenditure decisions using guesswork.

THE FIELD

Nameplate Field

Zero-friction data capture. Offline-first by design. Everything a technician needs to scan, inspect, and log service.



The Field captures
physical reality.
HQ commands
financial strategy.

HQ

Nameplate HQ

Total portfolio visibility. Live capex forecasting, warranty leakage tracking, and repair-vs-replace economics.



THE TAGGING PARADIGM

MANUFACTURER SERIALS

PROPRIETARY NAMEPLATE IDs (NPIDs)

Readability

Frequently corroded, painted over, or facing a wall (15-30% failure rate).

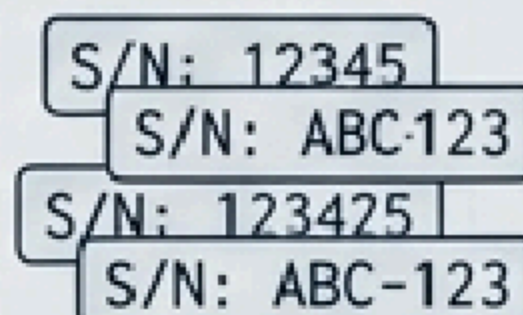


Error-corrected QR code, scans at arm's length in the dark.

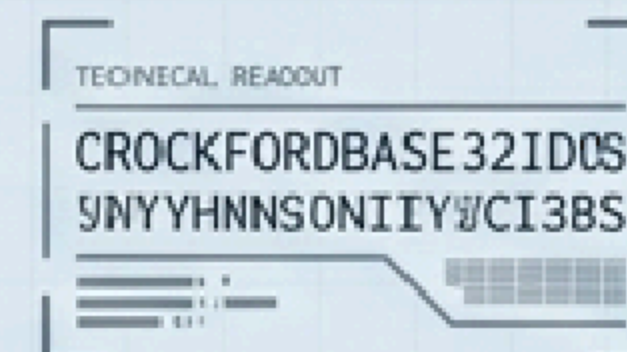


Uniqueness

Format overlaps across brands; not globally unique.



CROCKFORD BASE32 ID, MATHEMATICALLY GUARANTEED GLOBAL UNIQUENESS.



Pre-existence

Cannot be logged until found and transcribed.



Pre-allocated in-app. Minted in 3 seconds before the serial is even known.



Speed

45 seconds of hunting, flashlight, and error-prone typing.



Sub-3-second scan.



MANUFACTURER SERIALS ARE FOR INTELLIGENCE (WARRANTIES, PARTS). NPIDs ARE FOR OPERATIONS (TRACKING, SPEED, IDENTITY). WE CAPTURE BOTH, BUT WE BUILD THE LEDGER ON NPIDs.

THE NAMEPLATE TAG: ANATOMY OF AN ARTIFACT

[THE SUBSTRATE]

Tamper-evident destructible vinyl. Shreds upon removal, leaving visible evidence—the primary theft-deterrent property.

[THE ADHESIVE]

High-tack acrylic, rated for low surface energy and extreme temperature cycling (for water heaters and HVAC).

[THE PAYLOAD]

Error-corrected QR (Level H - 30%). Survives a torn corner, scratches, and lint-fuzz. Encodes a secure URL (np.app/a/...).

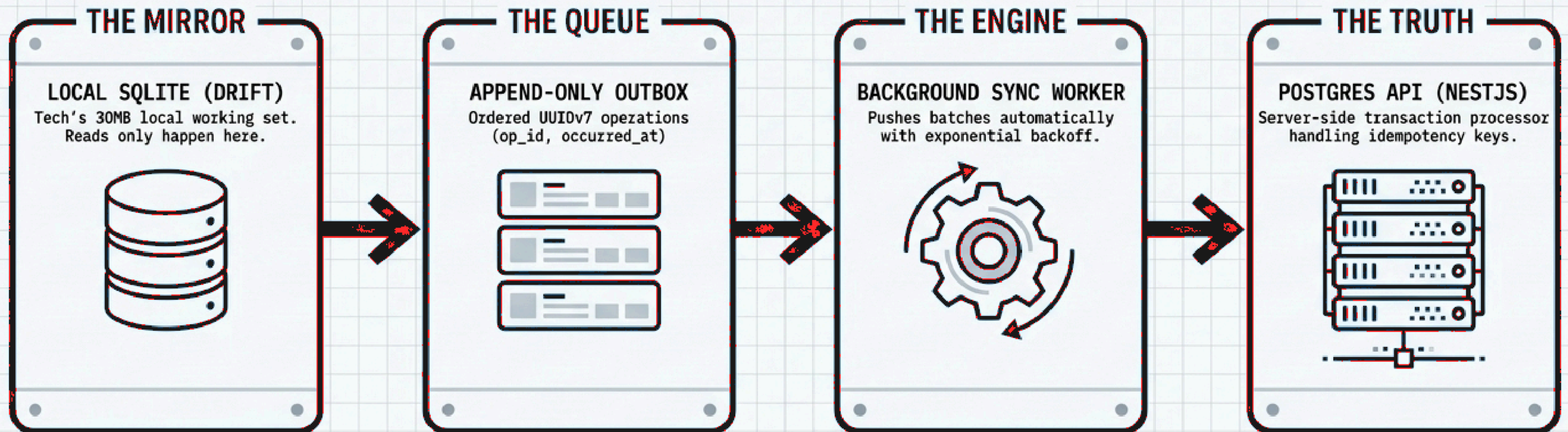
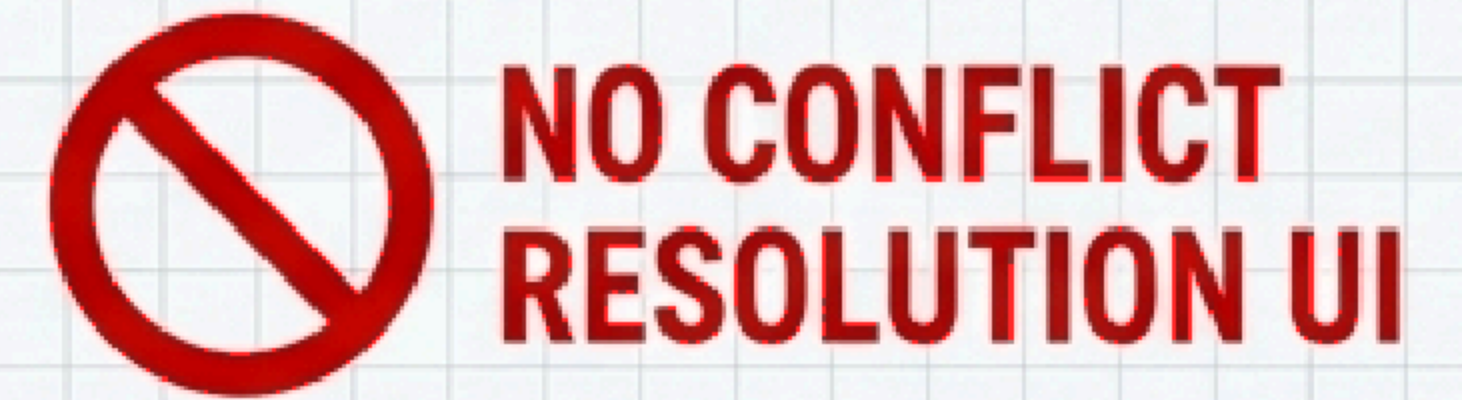
[HUMAN-READABLE KEY]

Unambiguous Crockford Base32 (no I, L, O, U) with a final check-digit to reject typos instantly.

NP-7K2M-4QX9

The artifact is engineered as rigorously as the software. Consistency is a speed feature: a tech should never have to hunt for the tag.

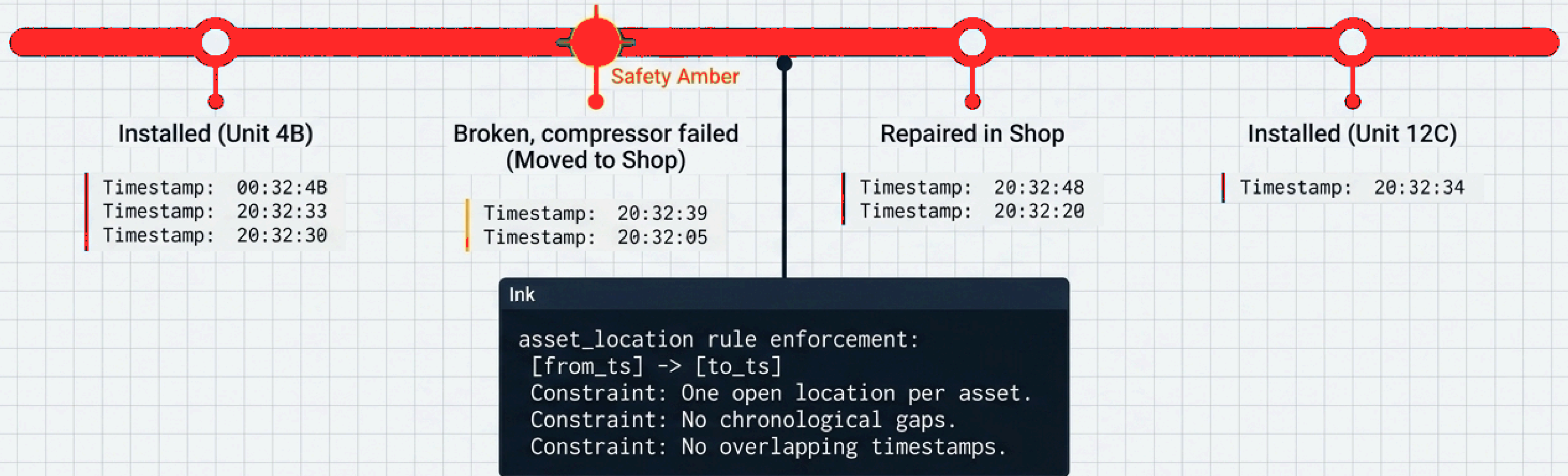
OFFLINE-FIRST BY DESIGN



A technician working a basement laundry room logs 11 service events and 40 photos with zero signal. When they walk into the parking lot, everything lands correctly. The app never blocks the tech to arbitrate state.

THE CHAIN OF CUSTODY (APPEND-ONLY LEDGER)

ASSET: NP-9P4T-2WB1



Mist

The Chain of Custody is strictly append-only. Location changes cannot be made via a silent field edit. If a mistake is made, it is corrected via an appended correction row. A manipulated history is visibly recorded in the history.

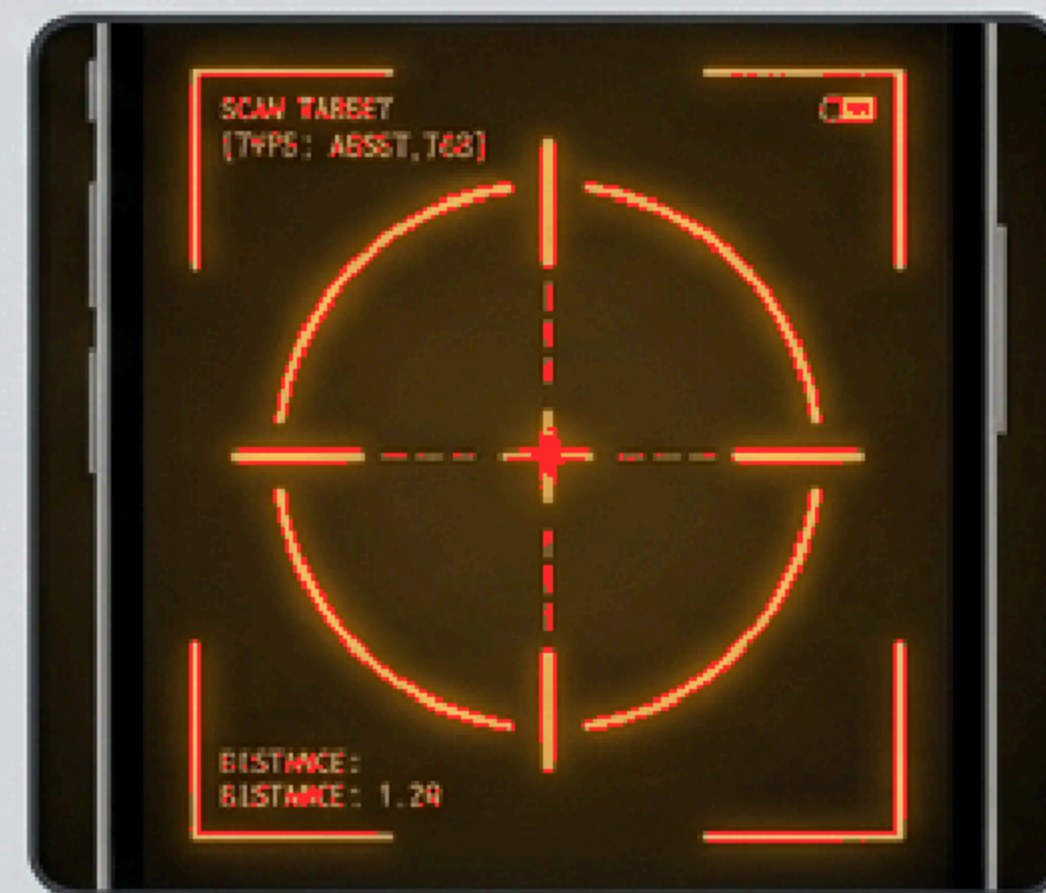
THE 4-MINUTE UNIT TURN

1. Roster Generation

A screenshot of a tablet displaying a checklist. At the top, it says 'CHECKLIST' with a checkbox. Below are three items: 'Refrigerator [S/R: 345X-980]' with a timestamp '15:58 AM' and date '15/18-2022, 13:55 AM'; 'Dishwasher [S/R: 210Y-456]' with a timestamp '28:58 AM' and date '15/18-2022, 13:35 AM'; and 'Microwave [S/R: 678Z-123]' with a timestamp '25:58 AM' and date '10/18-2022, 16:25 AM'. Each item has a checkbox to its left.

Server generates a dynamic checklist based on exact assets assigned to the specific unit.

2. Scan & Verify



Tech walks the room, scanning tags in seconds. Unrecognized assets get tagged inline.

3. Condition Triage

A screenshot of a triage interface with four red buttons arranged in a 2x2 grid. The top-left button says 'PRESENT [C_001]'. The top-right button says 'DAMAGED [C_002]'. The bottom-left button says 'NEEDS SERVICE [C_003]'. The bottom-right button says 'MISSING [C_004]'. Each button has a white border.

Fast, single-tap inputs for Present, Damaged, Needs Service, or Missing.

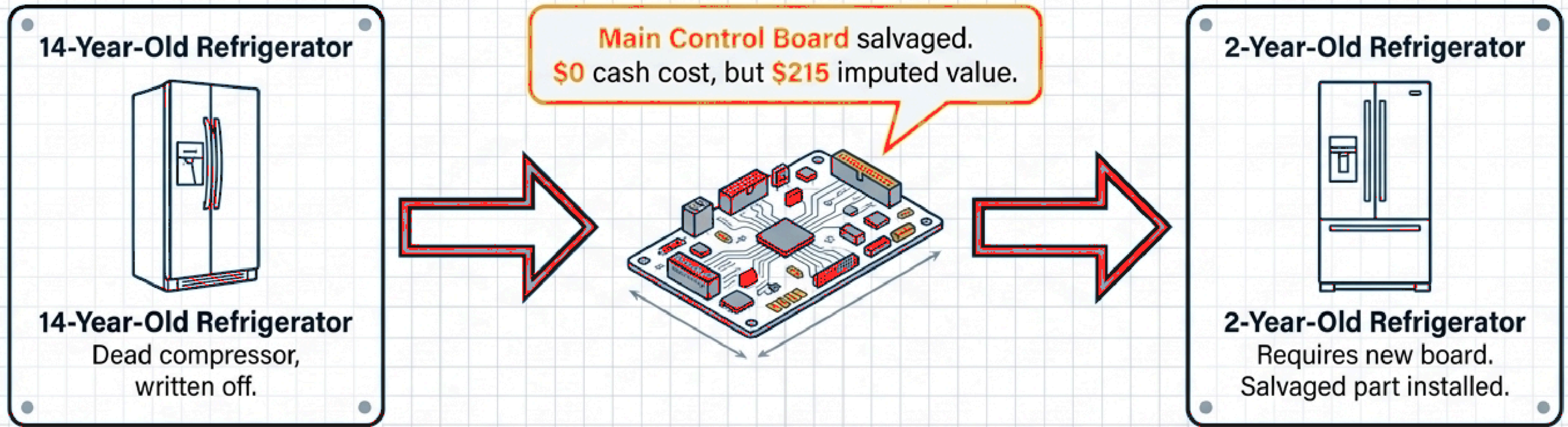
4. Auto-Emission

A screenshot of an auto-emission confirmation screen. It features a green checkmark icon at the top. Below it, the text reads: 'WALKTHROUGH COMPLETE. AUTO-GENERATING WORK ORDERS: [WO-1234, WO-5678]...'. At the bottom, it says 'SLA: 24HR'.

Completing the walkthrough automatically generates SLA-driven work orders for flagged items.

Walk the unit once. Create perfect certainty. Everything else is automatically written down.

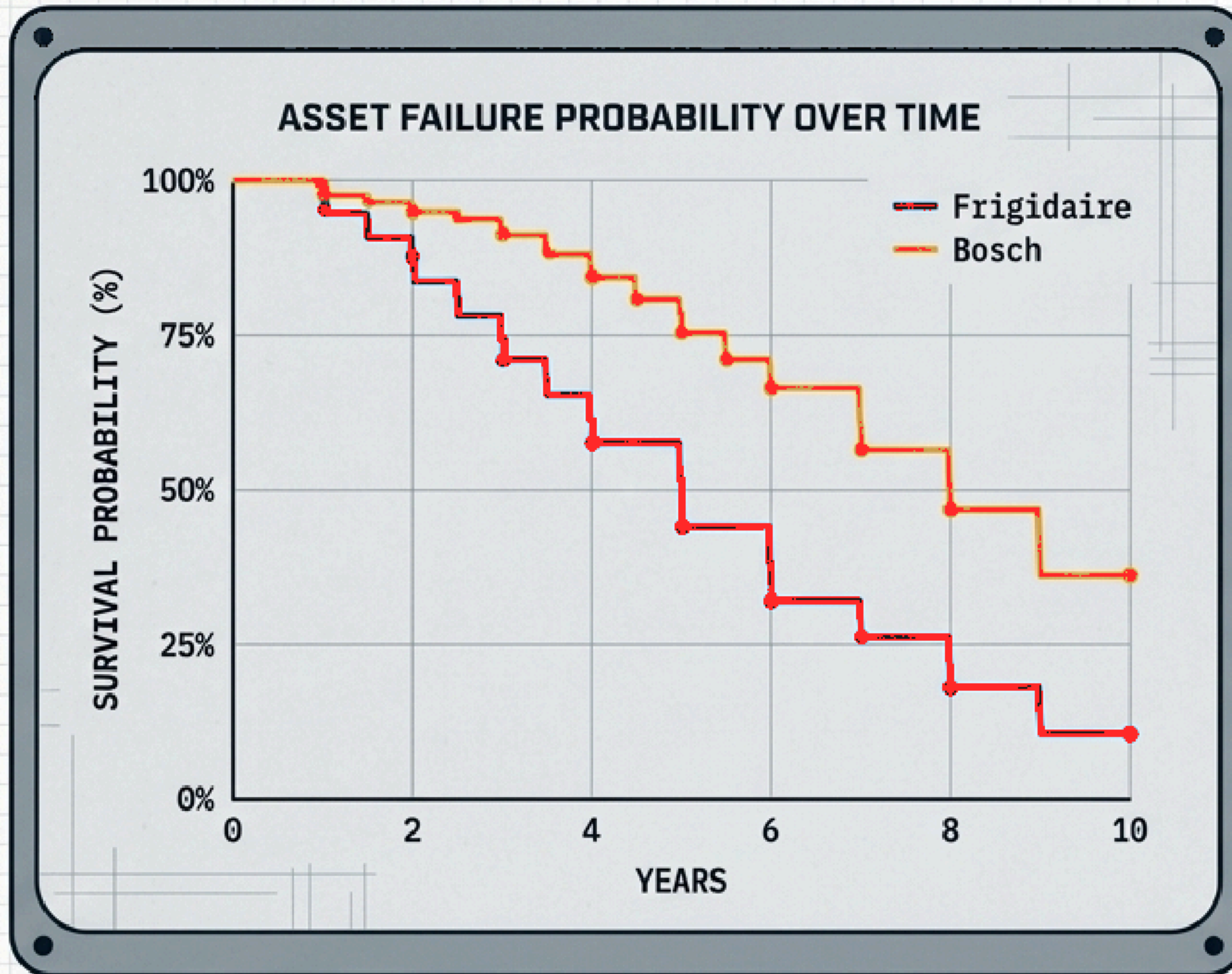
PARTS LINEAGE & HARVESTING



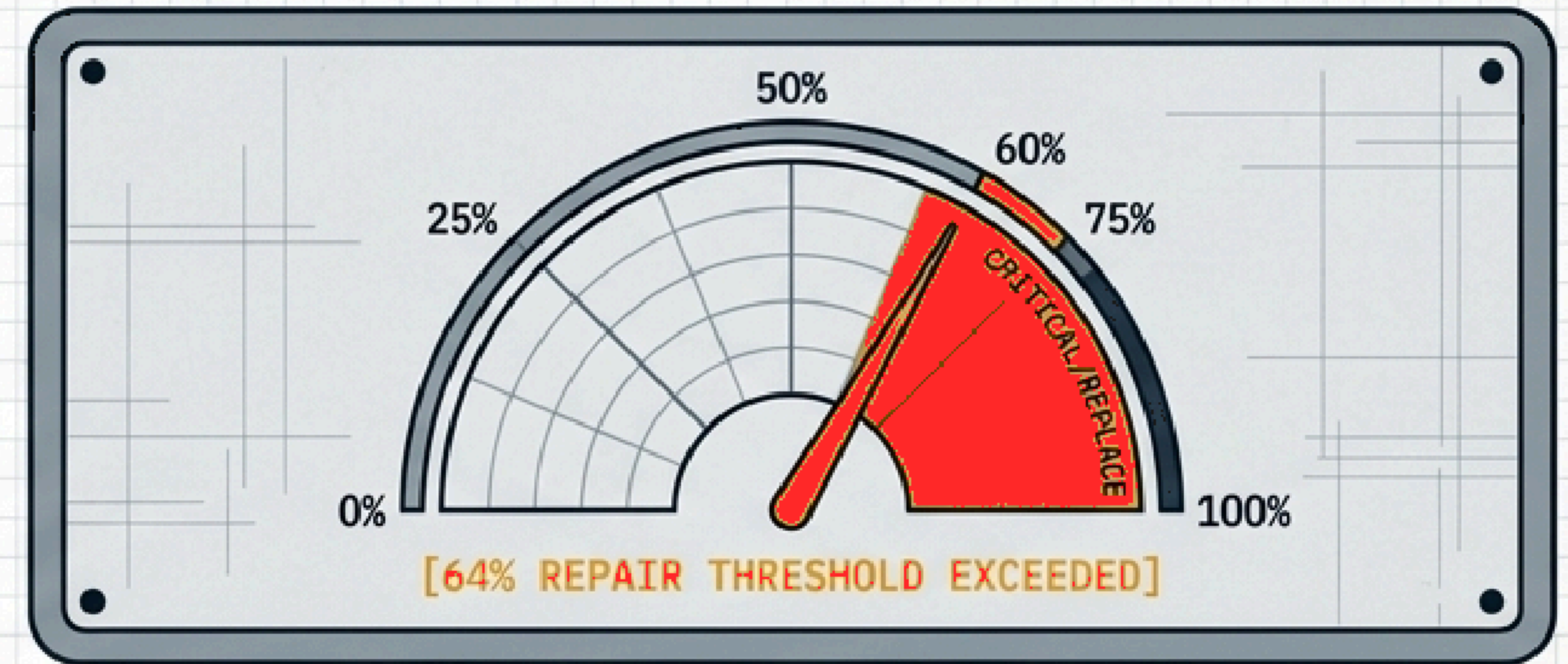
`part.source_asset_id` → `part.source_service_event_id` → `part_usage.installed_in_asset_id`

We don't just track line-item costs; we trace physical lineage. Nameplate makes the salvage discipline visible, rewarding shops for parts reuse and avoiding thousands in unnecessary purchases.

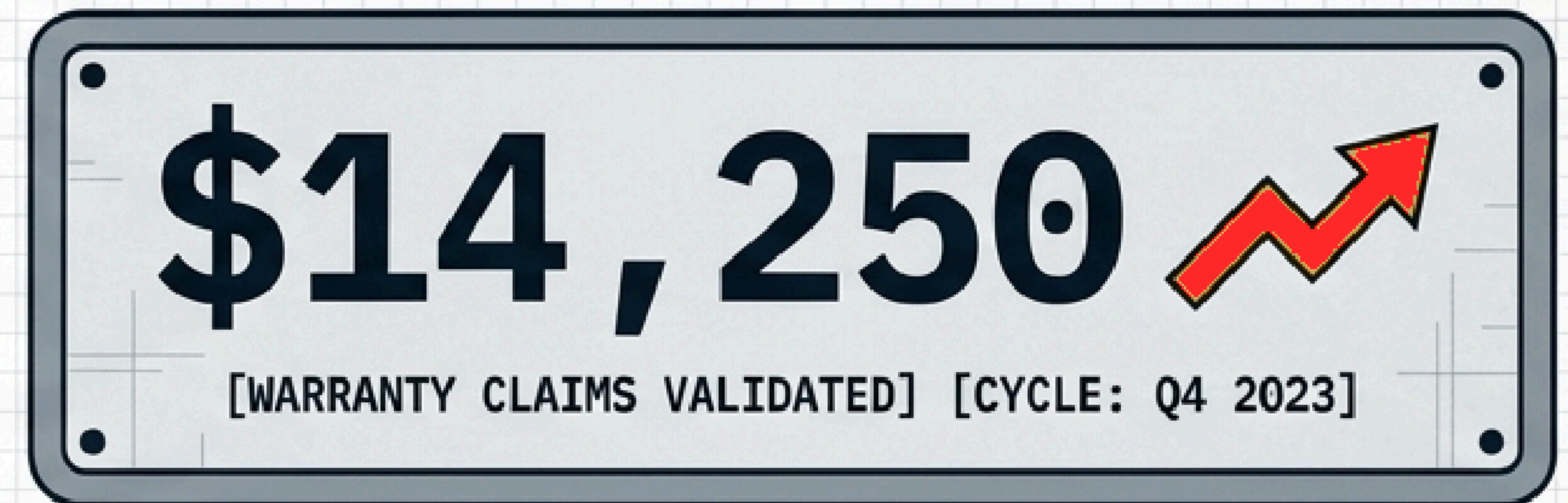
OPERATIONAL INTELLIGENCE AT HQ



Realized Lifespan & Capex Forecasting



Repair vs. Replace Economics



Dollars saved by intercepting out-of-pocket repairs on warrantied assets

The end goal isn't just knowing where things are. It's allowing your own numbers to dictate when to repair, when to replace, and what brand to buy next.

SHRINKAGE DETECTION

SYSTEM SIGNALS

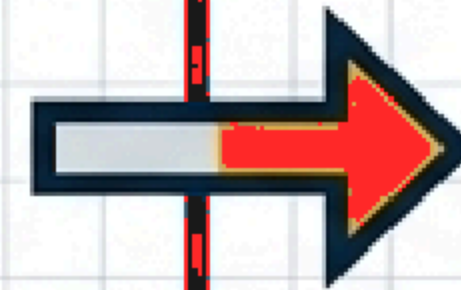
AUTOMATED ACTIONS

`current_location_confirmed_at > 180 days.`



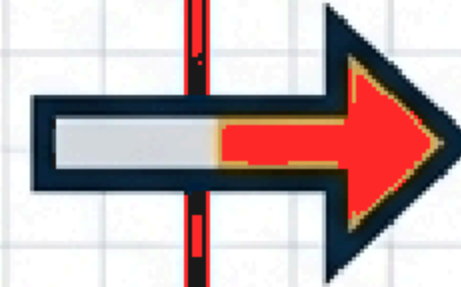
Triggers routine verification work order.

'Missing' finding during a unit turn.



Starts a **30-day Grace Window**.

`actual_property_id ≠ expected_property_id`



Logs an out-of-scope scan and flags for HQ review.

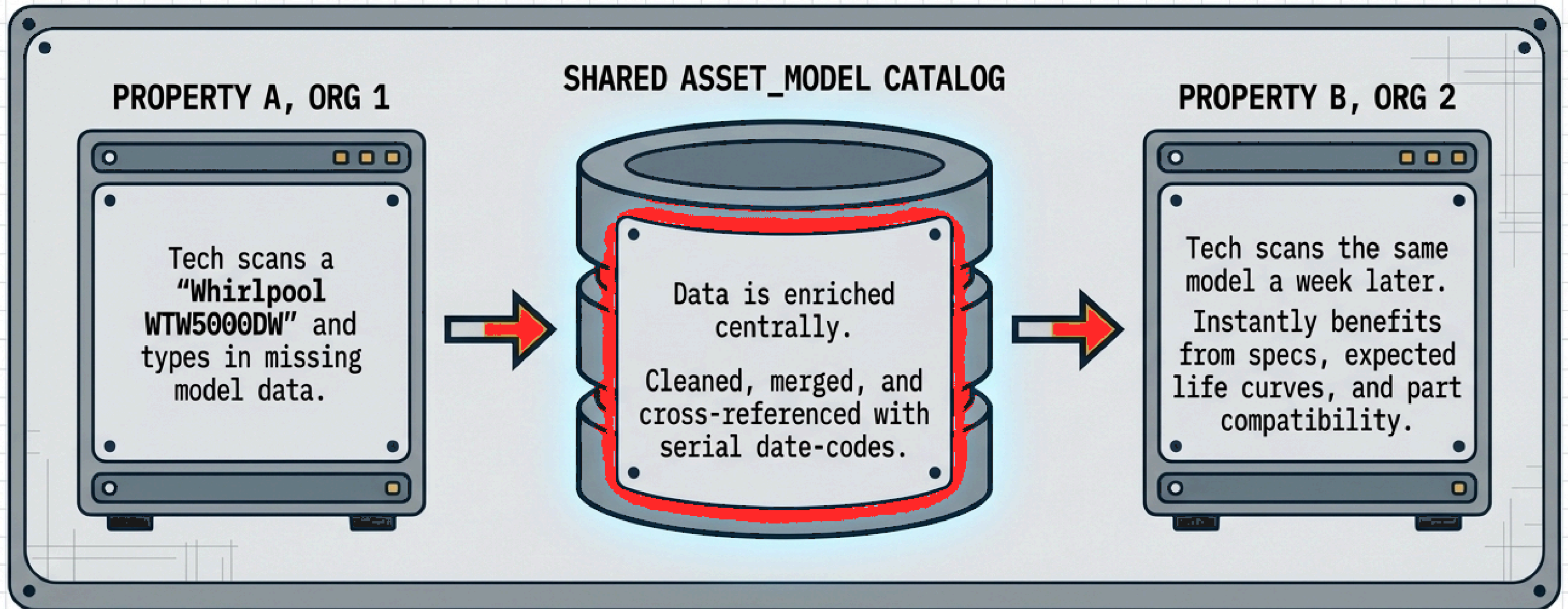
Grace Window expires with no scans.



Escalates to unaccounted-for status.

Shrinkage detection is about **assets**, **never people**. We never auto-accuse or rank technicians by loss. The 30-day grace window converts most 'missing' tags into benign relocations, preserving field trust. If techs feel **surveilled**, they stop scanning.

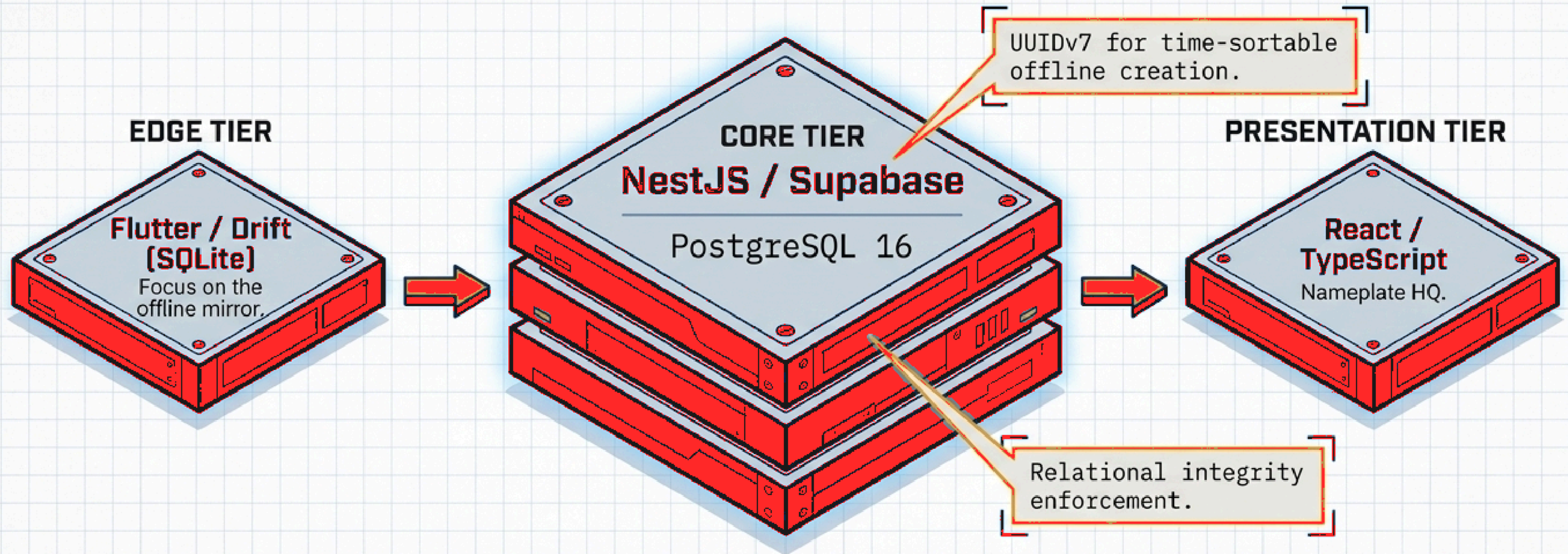
THE MOAT: CROWD-POPULATED CATALOG



No customer-identifying data is ever shared. Only manufacturer specs and aggregated lifespan statistics.

- After 50,000 tagged assets, Nameplate can definitively state: "The dishwashers in your portfolio fail at 6.2 years; comparables last 9.8." A data product built entirely from work users already do.

SYSTEM ARCHITECTURE (THE VO STACK)

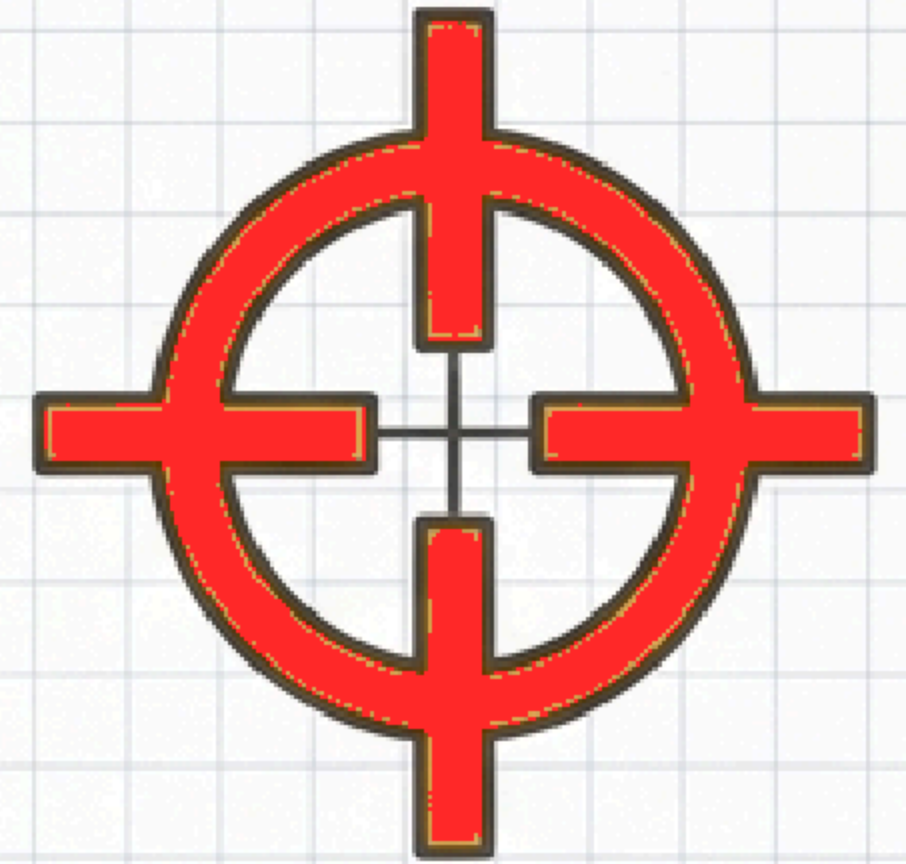


WHY NOT MONGODB OR FIRESTORE? BECAUSE HISTORICAL INTEGRITY IS THE PRODUCT.
The entity that matters most—part lineage across physical locations and time—is **graph-shaped**.
Relational constraints, window functions, and append-only ledgers cannot silently rot.

V0 Rollout & Success Metrics



Scan Verification Rate [Target $\geq 70\%$]



The percentage of service events and turn items backed by an actual hardware scan.

90-DAY PILOT PARAMETERS

- ✓ 200-500 units deployed.
- ✓ $\geq 95\%$ of in-unit assets tagged and categorized.
- ✓ $\geq 80\%$ of service events logged in-app rather than on paper.

If techs scan, the dataset compounds and every feature becomes possible. If they don't, no dashboard can save the product. Success in V0 is purely about frictionless field adoption.



**Walk the
unit once.**
Everything else
is already
written down.

Scan the plate.
See the history.
Log the fix.

Let's tag your portfolio.
nameplate.app/demo